

THE ROLE OF CROSS-FUNCTIONAL TEAMS ON THE ALIGNMENT BETWEEN TECHNOLOGICAL INNOVATION EFFECTIVENESS AND OPERATIONAL EFFECTIVENESS

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ABSTRACT

Organisations are increasingly investing in complex technological innovations, such as enterprise information systems, with the aim of improving the operation of the business, and in this way gaining competitive advantage. However, the implementation of technological innovations tends to have an excessive focus on either technological innovation effectiveness, or the resulting operational effectiveness. Focusing on either one of them is detrimental to long-term performance. Cross-functional teams have been used by many organisations as a way of involving expertise from different functional areas in the implementation of technologies. The role of boundary spanning actors is discussed as they bring a common language to the cross-functional teams. Multiple regression analysis has been used to identify the structural relationships and provide an explanation for the influence of cross-functional teams, technological innovation effectiveness and operational effectiveness in the continuous improvement of operational performance. The findings indicate that cross functional teams have an indirect influence on continuous improvement of operational performance through the alignment between technological innovation effectiveness and operational effectiveness.

Key words: Cross functional teams, technological innovation effectiveness, operational effectiveness, information systems alignment, continuous improvement, boundary spanning.

Track: Human resources issues in continuous innovation

1. INTRODUCTION

Innovative organisations are those that are able to use innovation to improve their processes or services (Tidd et al., 2001). These firms are faced with competitive pressures to improve efficiency and productivity through technological implementation (Ifandoudas and Chapman, 2006). Consequently, organisations that are seeking not only to survive, but to maximise operational effectiveness in an ever-changing environment, need to identify human resource factors, such as senior management support, training, employee empowerment and teamwork, as key elements in the implementation process of technological innovation (Beer and Eisenstat, 1996; Daily and Huang, 2001). Similarly, Organisations involved in innovative processes are increasingly using cross-functional teams to enhance their competitiveness (Pinto et al., 1993), and responsiveness to market changes (Tidd et al., 2001). Nevertheless, the extent to which these cross-functional project teams help organisations to increase the effectiveness of the operations and to continuously improve, after the implementation of technological innovations, has received little attention from researchers (Koulikoff-Souvion and Harrison, 2007). For instance, Harrison and Storey (1996) and Ahmad and Schroeder (2003) conducted studies to link human resources issues, such as cross-functional teams, with operations management and argued that operations management literature tends to either ignore or pay little attention to human resources issues, even given the critical role of people in achieving superior performance in the operations of service organisations.

As there are limited empirical studies which specifically investigated cross functional teams and their impact on the alignment between technological innovation and operational performance in service organisations, this research uses a quantitative approach to investigate the stated issue. This research addressed the question ‘What is the contribution of cross-functional teams to the continuous improvement of operational performance, after the implementation of technological innovations, such as enterprise information systems? This research analyses the results of a survey of employees in organisations from the electricity distribution and retail and higher education sector. The aim of this research is to build on the existing literature and to further confirm and refine a theoretical framework.

2. BACKGROUND

2.1 CROSS FUNCTIONAL TEAMS

In an ever-changing environment, superior leaders spend time creating the culture to support innovation, knowledge exchange and collaboration with the aim to improve processes or services (Moss, 2000; Tidd et al., 2001). In the quest to reach the improvement of processes or services, organisations implement technological innovations such enterprise information systems to improve the performance of their operations. In the implementation process of technological innovations, some organisations are increasingly using human resource factors such as cross-functional teams (Daily and Huang, 2001), because, cross-functional teams are often understood as the cooperation or collaboration of individuals drawn from various functional areas (Pinto et al., 1993). This cross-functional cooperation brings ideas, knowledge, expertise and innovation to the project. For instance, Tidd, et al (2001) stated that cross-functional teams have more to offer than individuals in terms of idea generation flexibility to solutions developed, and a mechanism for bridging boundaries to create innovative solutions within organisations. Likewise, a company with a strong culture of innovation, leverages relationships across boundaries and through the extended enterprise, and encourages people to reach further and faster to gain or spread knowledge (Moss, 2000; Robbins, 1997). Cross-functional teams provide a substantial range of ideas, learnings and improvements that can be applied to the organisation (Boer et al., 2001). Thus, organisations need to use cross-functional teams in the implementation of innovative technologies, such as enterprise information systems, with the aim of bringing ideas, learning, knowledge, expertise and innovation into the project. However, managers’ efforts to create complex networks to connect their people and partners globally and to empower employees for cross-functional teams to operate efficiently, could be seen as challenging if key factors to succeed are not appropriately defined.

According to McDonough (2000) many factors such as empowering project teams, establishing project climate, the human resources of the teams, and the establishment of goals have been suggested in the literature by a number of different researchers when looking at how to achieve successful teams. An appropriate project goal is mentioned most often as being associated with success (McDonough, 2000; Pinto et al., 1993). The establishment of clear and consistent goals provides the members of the cross-functional team with a common boundary or frame of reference and keeps the teams oriented towards common tasks and outcomes. This establishment of clear and consistent goals in turn promotes a higher level of cross-functional cooperation (Pinto et al., 1993). In addition, the team leader also may play an enabling role in fostering cooperation among team members, to keep the team focus on its goals (McDonough, 2000). Additional to the establishment of clear and consistent goals as a key factor in the successful organisation of cross-functional teams in the implementation of technological innovations, it is

also important to understand the expectations of stakeholders as they come from different functional areas and diverse cultures.

In regards to organisational cultural diversity, Orlikowski and Gash (1994) states that different stakeholders in an organisation have different assumptions, expectations, knowledge and perceptions of technological innovation. Such interpretations of technological innovation, called technological frames (Orlikowski and Gash, 1994) or three cultures of management (Schein, 1996), are central to understanding technological development, use, and change in organisations as they influence the way people behave towards technological innovations. These different stakeholders' expectations need to be analysed and taken into account while determining the project's goals and objectives as they influence the results of the project (Elias and Zwikael, 2007). Cross functional project teams are made of different stakeholders who bring a wide array of ideas, expectations and needs. Moreover, as the role of cross-functional project teams is important in effective technological implementation and responses of stakeholders such as managers, engineers, and operators-users, vary, care should be taken to minimise conflicts (Elias and Zwikael, 2007). Therefore, it is important to establish bridges that assist the members with different cultures or backgrounds in the cross-functional team, to succeed in the implementation of a common language so they can define clear and consistent goals.

Pitta and Franzak (1997) suggest that organisations involving external and internal cross-functional members create a boundary spanning team. Boundary spanning is tied closely with the management of technological, innovation, and strategic implementation to achieve organisational objectives. As stated by Schein (1996) different actors such as managers, engineers, and operators-users belong to different cultures, making it difficult to communicate with each other. Boundary spanning agents are viewed as communication stars (Tushman and Scanlan, 1981) and can effectively communicate widely within their work unit, across work units and outside their organisation. Kellog et al. (2006) argue that boundary spanning agents are able to act as translators, brokers or mediators. Cross-functional teams can reduce misunderstanding that arises in the different values found within functional areas if the cross-functional team members are able to develop a shared language and shared mental models. By doing this, actors from different functional areas can share information at all stages of the implementation of technological innovations such enterprise information systems, making it easier to identify problem areas early in the process and finding solutions that are shared by the team members in the same language.

2.2 OPERATIONAL EFFECTIVENESS AND CONTINUOUS INNOVATION

Bessant and Boer (2002) argue that organisations need to engage in continuous innovation, that is they need to be both operationally effective in exploitation and strategically flexible in exploration. It was often argued that these two capabilities could not be combined successfully. Nevertheless, Bessant and Boer (2002) maintain that developments in society, markets, technology and industry suggests that leading organisations need to find configurations of processes, procedures, people, technologies and organisational arrangements that allows them to become continuously innovative. According to Boer (2002) continuous innovation is the ongoing interaction between operations, incremental improvement, learning and radical innovation aimed at effectively combining operational effectiveness and strategic flexibility, exploitation and exploration. In seeking to develop a culture of continuous innovation there needs to be a focus on an organisation's capability to renew all or part of its managerial competencies and to create radically new competencies in order to achieve congruence with the changing business environment (Teece et al., 1997). In order to face changing market conditions, service firms need learning processes to build the flexible capability to reconfigure and transform their processes. In

dynamic and unstable environments firms need to constantly scan their environment and government policies, and develop agile behaviours or competencies to rapidly accomplish changes (Teece et al., 1997). In addition, an increasing number of factors are prompting organisations to seek to operate more efficiently and to ensure they have effective operational processes (Hill, 2000; Slack et al., 2004). This involves the need to deliver value-adding products or services of exceptional quality, on time, at a competitive price. Thus, organisations attempting to meet these objectives need to pay attention to their operational effectiveness as this is a primary driver of business performance (Slack et al., 2004).

Operational effectiveness refers to the ability to establish processes, based on core capabilities within the organisations, which work well (Porter, 1996). Operational effectiveness involves improving process performance by leading and controlling the processes within the firm as well as measuring and improving the processes. A better use of resources through these core processes enables the organisation to eliminate waste, and reduce costs, adapt more appropriate technological innovation and therefore perform better than competitors (Porter, 1996).

The five performance dimensions or objectives an organisation seeks to fulfil to attain operational effectiveness include cost, quality, flexibility, speed and reliability (Hill, 2005). Improving cost performance means that an organisation seeks the elimination of waste which comes from inefficiencies attained in processes such as purchasing, production, and staff performance. An appropriate disaggregation of the cost components impacting on the total cost performance of an organisation gives the opportunity to identify the areas for improvement (Slack et al., 2004). Furthermore, improving on quality provides an opportunity to bridge the gap between what organisations are capable of offering and what customers demand. That is, viewing quality as a consistent provision of services that satisfy customers rather than simply conforming to specifications without any clear continuous improvement. The third operational performance objective consists of being flexible, this includes an organisation's ability to adjust to changes in response to customers' needs (Slack, 1991). Additionally, improving on speed prompts an organisation to be able to shorten the time between the service request and delivery of the service, with the frequency, and at the time, that a customer requests (Hill, 2005). Finally, reliability suggests that an organisation's processes consistently perform as expected over time. That is, customers are satisfied by organisations that provide services that do not fail over a period of time or with services that are delivered as agreed (Porter, 1996).

2.3 OPERATIONAL EFFECTIVENESS AND BUSINESS PROCESS REENGINEERING

In the implementation of innovative technologies, managers need to identify specific sets of organisational capabilities to transform and reconfigure the organisation's core managerial competencies. Enterprise information systems implementation, organisational change projects, continuous improvement and business process re-engineering require different competencies and capabilities. All reconfigurations and transformations continuously affect operational processes by the accumulation, selection and change of patterns of routines and practices (March and Simon, 1993). The main difficulty with the implementation of innovative technology such as enterprise information systems is the dramatic change they bring to business operations. An enterprise information systems project is as much about changing the way a business operates as it is about technological innovation (Davenport, 2000).

Technological innovation should be seen as an enabler of business process reengineering, even if the extent of the technological change needed is great (Davenport, 2000; O'Neill and Sohal, 1998). In fact, the business process reengineering movement has largely been replaced by enterprise information systems initiatives (Davenport, 2000). So to establish a competitive

advantage over competitors depends on the ability of organisations to implement the technological innovation effectively and adapt their business processes to the requirements of the enterprise information systems as their competitors may have the same or an equally effective enterprise information systems project already well implemented and correctly aligned with the operations of the organisation. Business organisations can progress from continuous improvement to effective business innovation, by inculcating and developing a culture which encourages innovation. Thus, for this research business process reengineering is seen as a way to innovate, fundamentally impacted by the implementation of technological innovations such enterprise information systems.

2.4 TECHNOLOGICAL INNOVATION EFFECTIVENESS

Technological innovation effectiveness or system effectiveness can be described as the extent to which information systems contribute to achieving organisational goals and benefits (DeLone and McLean, 2003). Companies deriving the greatest benefits from their systems are those that, from the start, view them primarily in strategic and organisational terms. These companies stress the importance of operational effectiveness, not the system. However, the high failure rate in implementing such systems is a major concern (Davenport, 1998). Failure rates for large-scale system development projects are extremely high and many information system projects are failing to achieve their stated outcomes (Jamieson and Hyland, 2004). However, as it is difficult to quantify, the real level of information system failure could be far greater than is reported (Jamieson and Hyland, 2004). This prompts the need for a better understanding of the measures that assist managers in assessing the performance of an EIS through the evaluation of its dimensions.

The revised DeLone and McLean model (2003) includes six interrelated dimensions of information systems success: information quality, system quality, service quality, intention to use, user satisfaction, and organisational impact as dimensions to measure the dependent variable IS effectiveness. In the DeLone and McLean's success model, system quality measures technical efficacy – the desired characteristics of the system. This assessment is based on the performance and productivity of the system. Information quality is the measurement of output from EIS. It measures semantic success – characteristics of the information and its desired form; the degree to which information produced has the attributes of content, accuracy, and format required by the user. Service quality is the level of service received by the users of EIS and the manner in which the service is provided by the IS department as it influences the degree of satisfaction with an EIS. Use and user satisfaction measure effectiveness success through studies that attempt to analyse and measure the interaction of the information product with its recipients, the degree to which the user believes that using a particular system has enhanced his or her job performance. User satisfaction is defined as the user's response to the use of the output of an EIS, the psychological state after the use of an EIS. User satisfaction goes hand in hand with user involvement, particularly during the phases of the analysis, design and implementation of an enterprise information system in an organisation. In addition, Baroudi et al. (1986) argues that user involvement in information system development is generally considered an important mechanism for improving system quality and ensuring successful system implementation. Thus user involvement can be used as a dimension to measure system effectiveness.. Individual impact is the effect the information has on the behaviour of the user, including improving personal or departmental performance, relating to what influences the information product has on management decisions. This impact occurs when the information is received and understood by the users, and applied to their jobs.

In measuring performance it is important to have a clear understanding of the outcomes from the investment of a significant amount of human and economic resources in EIS solutions that cannot always be properly adapted to particular circumstances. Management accounting systems have been traditionally used to measure performance which focuses on data such as profit, return on investment and cash flow. These types of measures merely rely on financial performance and do not reflect the requirements that an organisation must fulfil in today's competitive business environment, or operational requirements. Thus, the effectiveness of the implementation of technological innovations such as enterprise information systems, should be measured in terms of the real operational benefits rather than through the achievement of information systems outcomes only. In this regard, the dimensions stated in the DeLone and McLean model (2003), in addition to the five performance objectives stemming from operational effectiveness are a good indication of the real effectiveness of a technological innovation implementation.

The main purpose of this research is to build on and extend the existing literature and to put forward a theoretical framework that examines the following three propositions:

1. That a limited number of factors have the potential to explain the alignment between technological innovation effectiveness and operational effectiveness; and
2. That, cross-functional project teams influence the alignment between technological innovation effectiveness and operational effectiveness.
3. That, cross-functional project teams influence the continuous improvement of the organisation.

3. RESEARCH ISSUES AND METHODOLOGY

This research is undertaken with an exploratory purpose as the alignment between system effectiveness, operational effectiveness and cross-functional teams and their influence in the continuous improvement of operational performance in service organisations has had little previous academic investigation. According to Sekaran (2003) exploratory study is undertaken when there is a lack of understanding of the problem which leads to an unstructured problem design. Quantitative data was gathered through a self administered questionnaire. The questionnaire was administered to managers, engineers (technologist), and administrative and operational staff as, according to Orlikowski and Gash (1994), different actors in an organisation have different assumptions, expectations, knowledge and perceptions of technological innovation. The questionnaire was also administered in organisations that had recently implemented an enterprise information system. In the process of constructing measures of key variables and refining the survey instrument, we developed four pilot tests that enabled us to introduce a number of revisions that were carried out to improve the survey instrument between the initial draft and the final instrument.

The questionnaires were emailed to employees in managerial or executive role, information technology / information systems or engineering role and operators or general staff in the selected service organisations. Of the 450 surveys distributed, 144 were returned (32% response). Each returned questionnaire was reviewed for completeness and, of the 144, 6 were considered unusable due to large amounts of missing data, lack of involvement of the respondent in the use of EIS, and the impossibility of identifying the role of the respondent (manager, engineer or operator-user).

4. RESULTS

4.1 PROPOSITION ONE

As the main purpose of the study is to explore the alignment between technological innovation effectiveness and operational effectiveness and the role played by cross-functional teams in the stated alignment, the next step in the data analysis is to perform a factor analysis to reduce the dimensionality and to identify the most important clusters from technological innovation effectiveness and operational effectiveness, while at the same time eliminating the items that are less representative. The factor analysis was performed using the maximum likelihood extraction method and oblique rotation method which allows the factors to be correlated. During the factor analysis process, four factors emerged. Hair et al (1998) argue that in order to ensure a power level of 80 percent, a factor loading of 0.55 is significant if the sample size is at least 100 observations at a significant level (α) of 0.05. Thus, only factor loadings of at least 0.60 have been considered. The solution resulted in a Kaizer-Meyer-Olkin (KMO) value of 0.832 with four factors accounting for 52.65% of the cumulative variance, indicating a satisfactory solution. The four factors are: quality and speed stemming from operational effectiveness and quality of information and quality of the service stemming from technological innovation effectiveness. These four factors also demonstrate the focus that organisations need to consider when aligning technological innovations with operational effectiveness.

4.2 PROPOSITION TWO

Forward multiple regression was used to assess the ability of cross-functional teams to predict the two dimensions stemming from technological innovation effectiveness and the two performance objectives stemming from operational effectiveness. The results from the multiple regression analysis (table 1) show that cross-functional teams is a significant predictor of the two dimensions stemming from technological innovation effectiveness and the two performance objectives stemming from operational effectiveness (Factor 1 'Operational Effectiveness Quality': $\beta=.32$; Factor 2 'Operational Effectiveness Speed': $\beta=.36$; Factor 3 'Technological Innovation Effectiveness Information Quality': $\beta=.35$; Factor 4 'Technological Innovation Effectiveness Service Quality': $\beta=.46$). Support is therefore found for proposition two, that, cross-functional project teams influence the alignment between technological innovation effectiveness and operational effectiveness.

Table 1. Multiple regression analysis
(TIE^γ: Technological Innovation Effectiveness; Inform^φ: Information;
OE^ψ: Operational Effectiveness)

Dependent Variable	Predictors	Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
TIE^γ Inform Quality	Cross-functional Teams	0.380	0.087	0.351	4.365	0.000
TIE^γ Service Quality	Cross-functional Teams	0.534	0.089	0.457	5.984	0.000
OE^ψ Quality	Cross-functional Teams	0.416	0.104	0.324	3.989	0.000
OE^ψ Speed	Cross-functional Teams	0.377	0.085	0.356	4.438	0.000

4.3 PROPOSITION THREE

Path analysis has assisted the researchers in describing the entire structure of linkages between independent and dependent variables (Ho, 2006). Table 2 represents entry of the set of predictors of Continuous Improvement of Operational Performance. The results show that Factor 1 ‘Operational Effectiveness Quality’ accounted for 57.1 % of the variance (R square) in the success of Continuous Improvement of Operational Performance. Likewise, Table 2 shows that Factor 3 ‘Technological Innovation Effectiveness Information Quality’ accounted for 15.2% of the variance (R square Change). Entry level of the independent variable Factor 2 ‘Operational Effectiveness Speed’, resulted in a significant F change, $F(1,134)= 6.10$, $p<0.05$ for Continuous Improvement of Operational Performance, and increased variance explained by 1.2% (R Square Change). The model as a whole explains 73.5% (R square) of the continuous improvement of operational performance. Factor 4 ‘Technological innovation effectiveness Service Quality’ and Cross-functional Teams have been deleted as the variance accounted (R square) less than 1.2% and a significance $p > .05$, and are not seen as predictors of continuous improvement of operational performance.

**Table 2. Predictors of continuous improvement of operational performance
(TIE^γ: Technological Innovation Effectiveness; OE^ψ: Operational Effectiveness)**

Dependent Variable	Predictors	R	R Square	Change Statistics				
Continuous Improvement of Operational Performance				R Square Change	F Change	df1	df2	Sig. F Change
	Factor 1 OE Quality	0.756	0.571	0.571	180.985	1	136	0.000
	Factor 3 TIE ^γ Information Quality	0.850	0.723	0.152	74.166	1	135	0.000
	Factor 2 OE ^ψ Speed	0.857	0.735	0.012	6.107	1	134	0.015

The path coefficients between continuous improvement of operational performance and the two dimensions stemming from technological innovation effectiveness, the two performance objectives stemming from operational effectiveness and cross-functional teams were obtained by regressing the former on the latter (Ho, 2006). The results from the coefficients table (see table 3) generated from the regression analysis show that only three dimensions or performance objectives entered the prediction equation as shown in figure 1 (Factor 1 ‘Operational Effectiveness Quality’: $\beta=.59$; Factor 3 ‘Technological Innovation Effectiveness Information Quality’: $\beta=.35$; Factor 2 ‘Operational Effectiveness Speed’: $\beta=.14$). Quality of the service stemming from technological innovation effectiveness and cross-functional teams has no predictive power or contribution on continuous improvement of operational performance.

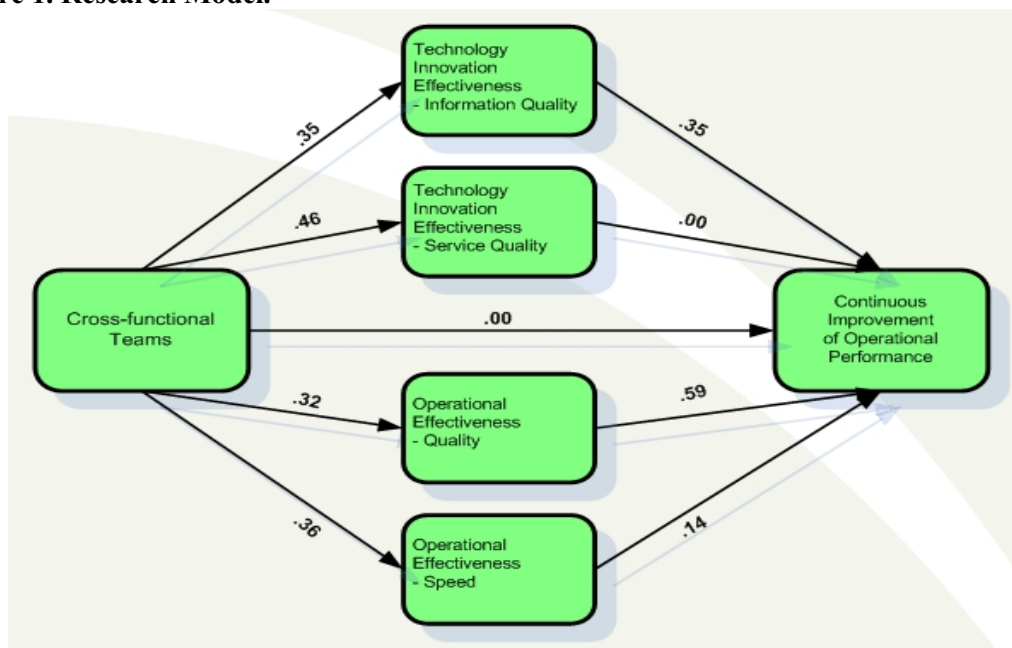
**Table 3. Path analysis for Continuous Improvement of Operational Performance
(TIE^γ: Technological Innovation Effectiveness; Inform^φ: Information;
OE^ψ: Operational Effectiveness)**

Dependent Variable	Predictors	Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
Continuous Improvement of Operational Performance	TIE ^γ Inform Quality	0.317	0.051	0.345	6.241	0.000
	TIE ^γ Service Quality	0.000	0.000	0.000	1.742	0.084
	OE ^ψ Quality	0.457	0.037	0.590	12.236	0.000
	OE ^ψ Speed	0.128	0.052	0.137	2.471	0.015
	Cross-functional Teams	0.000	0.000	0.000	-.161	0.873

Figure 1. Presents the path model together with the estimated regression coefficients (Beta values- β) associated with the hypothesized paths. The path analysis shows direct contribution from cross-functional teams on technological innovation effectiveness and operational effectiveness, however, there is no direct contribution from cross-functional teams on the continuous improvement of operational performance. Looking at the Coefficients table (Table 3),

the Beta weights (standardized regression coefficients) for the three factors, Factor 1 ‘Operational Effectiveness Quality’, has the strongest unique contribution to the prediction of Continuous Improvement of Operational Performance, whereas Factor 3 ‘System Effectiveness Information Quality’ and Factor 2 ‘Operational Effectiveness Speed’ are weaker. Furthermore, the positive coefficients indicate that the greater the focus on quality of the operational services, quality of information and quality of the operational speed, the more that organisation will see continuous improvements in the operational performance (Factor 1: $\beta=0.59$, $t=12.24$, $p<0.01$, Factor 3: $\beta=0.35$, $t=6.24$, $p<0.01$, Factor 2: $\beta=0.14$, $t=2.47$, $p<0.01$). The three independent variables are also making a significant contribution to the prediction of the dependent variable: Continuous Improvement of Operational Performance, where $p < 0.001$ for the three independent variables. Additionally, cross-functional teams plays an indirect role in the prediction of Continuous Improvement of Operational Performance, due to its significant influence on the two dimensions stemming from technological innovation effectiveness and the two performance objectives stemming from operational effectiveness (Factor 1: $\beta=0.32$, $t=3.99$, $p<0.01$, Factor 2: $\beta=0.36$, $t=4.44$, $p<0.01$, Factor 3: $\beta=0.35$, $t=4.37$, $p<0.01$, Factor 4: $\beta=0.46$, $t=5.99$, $p<0.01$). The positive coefficients indicate that the focus of cross-functional teams should be on operational objectives such as quality and speed and on the quality of the information as these three dimensions or performance objectives are important in the achievement of continuous improvement of operational performance. Support is therefore found for proposition three that, cross-functional project teams influence the continuous improvement of the organisation; however, the influence is an indirect relationship.

Figure 1. Research Model.



5. CONCLUSION

In answering the research question ‘What is the contribution of cross-functional teams to the continuous improvement of operational performance, after the implementation of technological innovations such as enterprise information systems?’ this study found that there is no direct influence of cross-functional teams on the continuous improvement of operational performance. The lack of direct influence of cross-functional teams on continuous improvement of operational performance can be explained by the fact that improved operational performance is the result of

the interaction between cross-functional teams and its related outcomes, i.e. quality of the information, quality and speed of the operations. Likewise, this study has demonstrated that the cross-functional teams have an important role in both the effectiveness of the operations and the effectiveness of the implemented innovative technology. Consequently, operational performance can be improved by the cross-functional teams but only via the effectiveness of operations and the effectiveness of the technological innovation.

This research has demonstrated the importance of including the performance objectives from operational effectiveness, in the context of information systems, to have a real understanding of the effectiveness of the implementation of technological innovations such as enterprise information systems. The findings from this study suggest that there is a strong relationship between cross-functional teams and operational effectiveness as well as between operational effectiveness and continuous improvement of operational performance. This indicates that when formulating strategies it is not only important to put an emphasis on system effectiveness but also on the incorporation of operational performance objectives in order to maximise the improvement in operational performance. Accordingly, technological innovations should be focused on attaining the needs of the operations in the performance objectives: quality, cost, speed, flexibility, and reliability. However, special attention must be given to the quality and speed performance objectives as this study has demonstrated that these two are the most important factors in achieving operational effectiveness.

This research has also shown that the goals for cross-functional teams to operate efficiently are: quality of the information, quality of the service and quality and speed of the operations. If these goals are clear, consistent, correctly formulated, implemented and monitored, the members of the cross-functional team will keep oriented towards common task outcomes. However, it is important to highlight that the members of the team must have the same understanding of these goals, as different interpretations, different mental models and different meanings can put the effectiveness of the cross-functional team in jeopardy. Conversely, it is important to stress the role of the boundary spanning agents, as they are the members of the cross-functional team in charge of translating and sorting the language of managers, engineers and operators or general users, so the information is meaningful to all members of the cross-functional team.

Finally, this research supports McDonough's (2000) conclusion, that the use of cross-functional teams is significantly associated with performance, indicating that employing cross-functional teams does have a positive impact on project performance. However, we expand this argument giving clarity to the goals that cross-functional teams must have to achieve success. Pinto et al. (1993) argue that establishing goals can impact on project performance, but only indirectly by affecting cross-functional cooperation. This research has demonstrated that without the proper intervention of boundary spanning agents, cross-functional teams are more likely to fail due to lack of clear and consistent goals.

It is important to note that decisions about innovation are made based on evidence, so organisations need high-quality information. One of the problems in continuously innovating organisations is that although they implement an EIS, this does not lead to improved operational effectiveness. The organisations studied are becoming more complex and more dynamic and they are seeking to innovate to deliver high-quality services, cheaper and faster. However, the extent to which this innovation helps organisations in the delivery of better services and in reducing operational cost is questioned by this study. The challenge for managers is to develop and adopt a prevailing set of goals and set of operating rules and procedures for the enhancement of quality across the organisation, in an effort to promote higher levels of improvement in the performance of operations. Nevertheless, achieving higher levels of quality appears to be more complicated

than previously thought, as employees at different levels must have a real commitment to achieve quality in their day-to-day operations. This suggests that organisations must work on all the dimensions mentioned in this research, to maximize the probability of achieving cross-functional team success and avoid the conflicts among different technological frames. The challenge for these organisations is to improve the quality of the information, the quality of services and the speed to deliver high-quality services to the end customers, as the research has revealed the importance of these dimensions or performance objectives in achieving the continuous improvement of operational performance.

6. LIMITATIONS AND FURTHER RESEARCH

The researched organisations achieved performance improvements through the utilisation of cross-functional teams. This improvement was indirectly obtained thanks to the alignment between technological innovation effectiveness and operational effectiveness. It would be appropriate then to conduct research to establish whether the same improvements can be attained without the participation of cross-functional teams. It is important to develop a qualitative stage to confirm the finding from this quantitative data analysis.

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